

SHARAD Subsurface Interface Report

Product: r_0277201_001_ss19_700_a

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Detection Summary

Confidence: HIGH
Detection rate: 60.4% of traces
Coherent segments: 290
Mean interface depth: 20 range bins (0.74 us two-way)

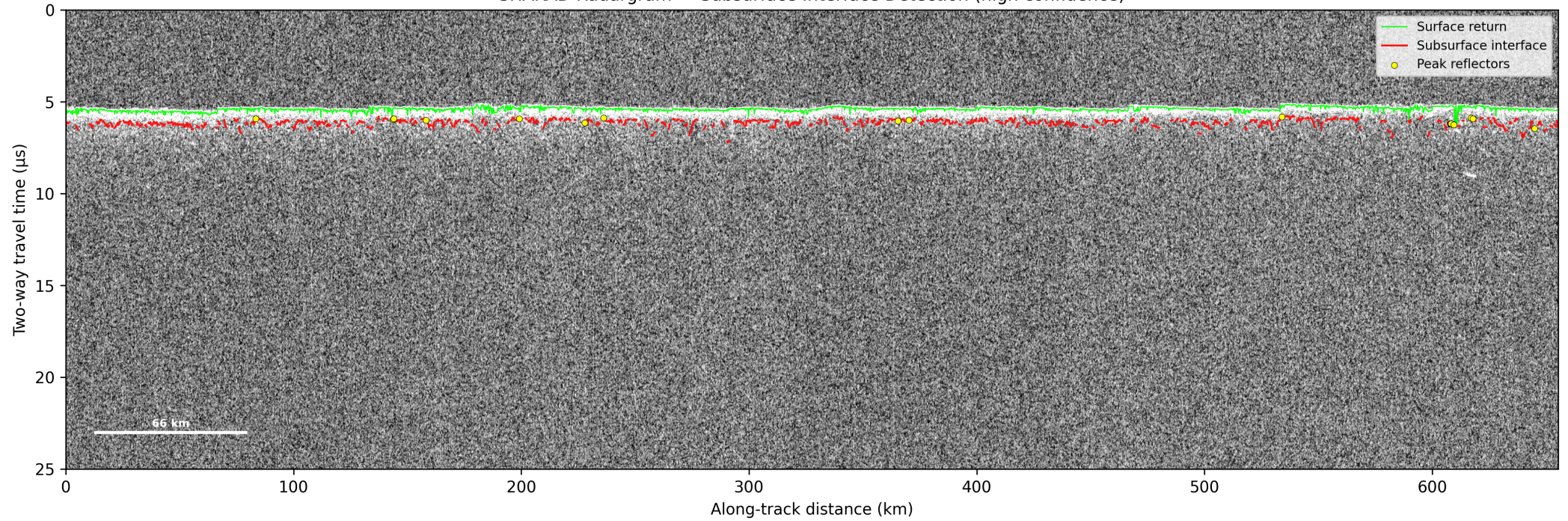
Depth Estimates

Material	er	Mean (m)	Min (m)	Max (m)
Vacuum	1.0	111.4	73.1	264.2
Porous Ice	2.8	66.6	43.7	157.9
Pure Water Ice	3.1	63.3	41.5	150.1
Basalt (low)	4.0	55.7	36.5	132.1
Basalt (mid)	5.0	49.8	32.7	118.2
Basalt (high)	6.0	45.5	29.8	107.9

Note: No co-located HiRISE DTM available. Cross-section comparison skipped.

Annotated Radargram with Subsurface Interface

SHARAD Radargram — Subsurface Interface Detection (high confidence)



Interface Detection Mask



Scientific Interpretation

A laterally coherent subsurface reflector was detected with HIGH confidence.

Assuming pure water ice ($\epsilon_r = 3.1$), the mean interface depth is 63.3 m (range: 41.5-150.1 m).

Assuming dry basaltic regolith ($\epsilon_r = 4.0$), the mean depth is 55.7 m.

Dielectric back-calculation was not possible (insufficient DTM coverage or relief).